

first name

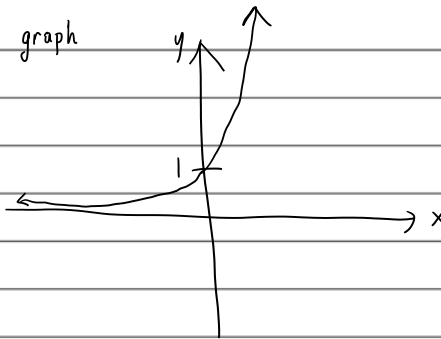
last name

email

In all quizzes and exams, show your work and write with correct notation.

(10pts) 1. Find $\lim_{x \rightarrow -\infty} e^x = \bigcirc$

due to the graph

(10pts) 2. Find a horizontal asymptote of $f(x) = \frac{1}{x^6}$

$$\lim_{x \rightarrow \infty} \frac{1}{x^6} = 0 \quad \text{so} \quad y=0 \quad \text{is a horizontal asymptote.}$$

★ In calculus, given a function,

Sometimes you have to find

 $\lim_{x \rightarrow \infty} f(x)$ and sometimesyou have to find $f'(x)$ This works but is longer. Note this is this

Alternative answer: $\lim_{x \rightarrow \infty} \frac{1}{x^6} = \lim_{x \rightarrow \infty} \frac{1 \cdot \frac{1}{x^6}}{x^6 \cdot \frac{1}{x^6}} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x^6}}{1} = \frac{\lim_{x \rightarrow \infty} \frac{1}{x^6}}{1} = \frac{0}{1} = 0$ so $y=0$ is a horizontal asymptote

Split is allowed because not zero

Side $\lim_{x \rightarrow \infty} \frac{1}{x^6} = 0 \quad \lim_{x \rightarrow \infty} 1 = 1$

(10pts) 3. If $f(x) = 7^x$, find $f'(x)$ with derivative rules.

$$f'(x) = 7^x \ln(7)$$

Need 7 here, not x
not 7^x

☹ $f(x) = 7^x \ln(7)$

☹ $7^x = 7^x \ln(7)$

Derivative rules are allowed.

Doing $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{7^{x+h} - 7^x}{h}$$

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has messy/challenging algebra